

Bitana Research Plan

Last updated: Sun 2026-08-16. Live config: v1.1.1 (*bf434ed*; Asia neutral-only + NY h16-17 pilot uncommitted). Register expanded to max 10. Companion prompt for deep research sessions: *research/QUANT_RESEARCH_PROMPT.md*. Weekend block artifacts: *research/output/reports/stop_variant_first_read_2026-08-01.csv*, *hmm_oos_validation_2026-08-01.csv*, *hmm_oos_decision_2026-08-01.json*, *weekend_aug1_bundle.json*. Runner: *research/weekend_aug1_analysis.py*. Single source of truth for what we test, when, and how results get promoted into (or killed out of) the live book.

Ground rules

- Objective: understanding > backtest performance. Every apparent edge is presumed false until it survives out-of-sample.
- Standard evaluation: **candidate-specific** cap-3 portfolio sim, 1 per symbol, 12 bps round-trip costs, PnL in ATR (1 ATR $\approx$ 0.8% equity at 8% risk / 10 ATR stop).
- No lookahead: regime and HMM labels from completed 4h bars only; models frozen before their evaluation window (HMM walk-forward: train $\leq$ month-end, filter forward).
- Pre-registration: hypothesis, mechanism, test window, and pass/fail criteria are written in this file BEFORE outcomes are computed.
- Selection windows never overlap evaluation windows.
- Sample floors: no conclusion on < 15 cap-3 accepted trades or < 5 distinct days (regime-split cells: $\geq$ 3 days). Top day may contribute $\leq$ 40% of net.
- Max 10 active hypotheses at once. Live config changes ship at week boundaries (Mondays), except safety cuts.
- Research integrity:** do not use the shared historical *would_live_accept* column for promotion decisions on trades opened before the Jul 23 fix. From Jul 23 onward, *would_live_accept* is strategy-scoped (only that strategy's previously accepted opens count). Offline analyses must still recompute candidate-specific acceptance when mixing pre/post windows.

Promotion ladder

Gate	Meaning	Objective criteria
G0	Idea	Pre-registered: claim, mechanism, window, criteria
G1	In-sample candidate	cap-3 net/trade $\geq$ +0.5 ATR, $n \geq 20$, ≥ 5 days
G2	OOS-validated	≥ 1 week AND ≥ 15 accepted OOS; net/trade $\geq$ +0.5 (volume-cutting filters: $\geq$ +1.0); concentration + cost checks pass
G3	Live pilot	1-2 weeks at 4% risk; slippage $\leq 1.5\times$ modeled; live/shadow fill agreement $\geq 90\%$
G4	Full size	8% risk

Kill at any gate: OOS net/trade < +0.2 ATR after 30 accepted, mechanism disproven, or day-concentration fails. Killed ideas go to the decision log below.

Dated timeline

Week of Jul 20 (go-live week)

- Mon Jul 20 — go-live day 1.** Fund check, first-fill reconciliation, live vs shadow *would_live_accept* agreement, age throttle armed (4% if neutral > 64h).
- Sun Jul 19 —** 4/6/8 ATR stop variants deployed to SHADOW_STRATEGIES (logging only): *asia_pump_short_4h_s{4,6,8}*, *ny_flush_buy_4h_open_s{4,6,8}*.
- Wed Jul 22 — live session retune shipped.** Asia remains *asia_pump_short_4h* at 48 bars, neutral+bull only, weekends off. NY switches to full-session *ny_flush_buy_4h*, neutral+bear only, Mon/Sat/Sun off. Blind Asia 36-bar cut rejected (MFE peak $\neq$ executable exit).

- **Thu Jul 23 — live fill accounting fix + weekly cooldown removed.** Binance market fills without avgPrice/cumQuote now resolve via /fapi/v1/userTrades. Corrected Asia closes: ETH -0.148R, SOL +0.094R (near flat). Weekly loss/cooldown path deleted from runtime. Live/shadow fill integrity returns to daily monitoring.
- **Thu Jul 23 — research instrumentation.** Candidate-specific cap-3 would_live_accept; full-session NY stop ladder ny_flush_buy_4h_s{4,6,8} added to shadow (pairs with live book, not open-window).
- **Fri Jul 24 - Sat Jul 25 — HMM OOS validation #1** (spec below). Not computed that weekend; executed Sat Aug 1 with frozen train $\leq$ Jun 30.
- **Sun Jul 26 — weekly review #1** (loop below).

HMM OOS validation #1 (pre-registered Jul 18) — RESULT Aug 1: KILL

- Frozen model: 6-state HMM trained $\leq$ Jun 30. **No retrain before this test.** ✓ held.
- Primary hypothesis: Asia entries during HMM states {H2, H5} outperform; other states are skippable. Filter was selected on Jul 1-17 data.
- Evaluation window: Jul 18 00:00 → Jul 24 24:00 UTC (fully out-of-sample).
- Metrics: (a) net/trade of in-state cap-3 Asia trades; (b) net PnL of trades the gate would have blocked; (c) mean filtered confidence on traded bars.
- **PASS** → G3 pilot (Asia HMM gate at 4% risk): in-state net/trade $\geq +1.0$ AND blocked net ≤ 0 AND confidence $\geq 70\%$, on ≥ 15 accepted.
- **EXTEND** one week: < 15 accepted in-state, or mixed.
- **KILL**: in-state net/trade $< +0.2$ on ≥ 15 accepted, or blocked trades netted $> +10$ ATR.
- Secondary (report only): H5-only variant; confirm NY H4/H5 still shows no benefit.
- Scope note: live Asia book trades neutral+bull without HMM gate. This test was incremental value of H2/H5 only.

Computed Aug 1 (live-like Asia filter: weekday + neutral/bull; candidate cap-3; 12 bps):

Window	Filter	in_cap3	days	avg_net	sum_net	blocked_sum	conf	Verdict
primary Jul18-24	H2+H5	11	3	-2.764	-30.40	-4.10	85%	EXTEND (n<15)
extended Jul18-31	H2+H5	15	5	-1.915	-28.73	-3.22	83%	KILL (avg $< +0.2$ on $n \geq 15$)
post-test Jul23-31	H2+H5	6	3	-0.410	-2.46	+1.95	66%	thin / no rescue
Jul1-17 selection (ref)	H2+H5	37	—	+1.808	+66.91	—	—	in-sample only

- Day driver on OOS: **2026-07-21 ≈ -30.6 ATR** dominates H5 bucket (top-day share $\gg 40\%$).
- H5-only mirrors H2+H5 (H2 almost empty OOS). NY H4+H5 secondary: still no benefit (cap3=3, avg ≈ -1.06 on both full-session and open books).
- **Decision: KILL Asia HMM {H2,H5} gate for live.** Selection-window edge did not transfer. Do not ship G3 pilot. Keep frozen Jun30 model only as research reference.
- **Aug 1 retrain: DEFER for gate use.** No live HMM path. Optional research-only train $\leq$ Jul31 snapshot may be versioned later; must not feed live gates without a new pre-registered OOS protocol.
- Artifacts: research/output/reports/hmm_oos_*_2026-08-01.*

Week of Jul 27

- Semi-Markov age throttle validation #1: did $> 64h$ -neutral trades underperform as predicted, live and shadow?
- Asia partial-profit rule (take 50% when $\geq +2$ ATR at 1h) OOS check on Jul 18+ data — measurement only; not an active promotion hypothesis.
- London/Late: begin OOS clock only if continuous quality-floor fills appear; instrumentation is already active.
- **Sat Aug 1 — HMM monthly retrain deferred** (see KILL above). OOS #1 executed.

August

- **Sat Aug 1 / Sun Aug 2 — stop-variant first read DONE** (Asia s4/s6/s8 from Jul 20; full-session NY s4/s6/s8 from Jul 23). ≥ 5 distinct Asia days (7 live-like); NY live-like days = 4 → **NY day floor not met** (extend to Aug 9+). Note: tighter stop without cutting risk_pct increases size — path R can rise while USD DD worsens.
- **Sun Aug 9 —** (a) NY stop-ladder re-read if ≥ 5 live-like days; (b) earliest London/Late promotion only if ≥ 20 fresh OOS accepted with candidate-specific cap-3 + concentration checks.

- **Sun Aug 16 — weekend read #1** (measurement; live Asia/NY weekends remain off).
- **Sun Aug 16-23** — NY scale-in decision (measurement/backlog; needs ≥ 30 samples).
- **Sun Aug 30** — stop-variant decision: replace 10 ATR only if a variant clears G2.

Stop-variant first read (Aug 1) — RESULT: no live change

Method: candidate-specific cap-3, 12 bps, live-like session filters (Asia: weekday + neutral/bull; NY: Tue-Fri + neutral/bear). Paired delta uses baseline cap-3 keys.

Asia ($\geq 2026-07-20$, live-like)

Strategy	raw	filt	cap3	days	avg_net	sum_net	PF	stops	paired Δ avg vs 10ATR
asia_pump_short_4h (10ATR)	61	43	26	7	-1.229	-31.95	0.51	3	—
s4	77	52	35	7	-0.641	-22.43	0.65	12	+0.378
s6	66	48	30	7	-0.668	-20.03	0.66	7	-0.150
s8	64	46	29	7	-1.011	-29.32	0.56	5	-0.006

NY full-session ($\geq 2026-07-23$, live-like)

Strategy	raw	filt	cap3	days	avg_net	sum_net	PF	stops	paired Δ avg vs 10ATR
ny_flush_buy_4h (10ATR)	115	73	23	4	-0.759	-17.45	0.56	1	—
s4	130	80	30	4	-0.330	-9.91	0.78	8	+0.208
s6	121	76	28	4	-0.710	-19.89	0.61	6	-0.228
s8	115	73	24	4	-0.879	-21.09	0.52	3	+0.129

Notes: - Entire live-like window is **red for all books** (regime/period effect); ranking is relative only. Do not promote on absolute edge. - **s4** is the only consistent relative improve (Asia +0.38, NY +0.21 paired avg) via cutting fat tails — at cost of many more stop hits (Asia 12 vs 3; NY 8 vs 1). - **s6/s8** do not beat baseline on paired avg; s6 Asia slightly worse. - Concentration: NY top day 2026-07-28 $\approx +12.6$ ATR (~ 60 -127% of book net) — fragile. - Asia symbol skew: ETH dominates losses (10ATR cap3 ETH $n=13$ $\text{avg} \approx -1.84$). - G1 not cleared (need cap-3 net/trade $\geq +0.5$). **Keep live stops at 10 ATR.** - Next: continue shadow through **Aug 30 G2 decision**; interim NY day-count check Aug 9. If s4 stays best relative but books stay negative, still no live swap (robustness preference: capped stop only when edge is positive OOS).

September

- **Sun Sep 6 — weekend verdict** for Asia at ± 1.0 ATR resolution (measurement).
- Event-driven — bull-book formal G2: Asia plain short in the current bull window (provisional live already) needs multi-day stable OOS vs shadow.

Monthly (first weekend of month)

- HMM walk-forward retrain + state-profile drift check.
- Session $\times$ regime matrix refresh; decision-log audit.
- Regime-detector audit report (flip lag / false transitions) from `storage/v6_telemetry.db` regime snapshots — measurement, not threshold tuning on the same window.

Weekly research loop (Sundays, $\sim 1h$)

1. **Integrity:** insert errors, enriched-field null rates, orphaned rows; live vs shadow reconciliation (fill agreement, slippage vs 12 bps, strategy-scoped `would_live_accept` on post-Jul-23 rows).
2. **Variance scan:** PnL by regime $\times$ age-bin $\times$ session $\times$ HMM state $\times$ weekday. Flag cells with $|\text{avg}| \geq 0.5$ ATR and $n \geq 15$ not explained by known axes.
3. **Register update:** advance / extend / kill each active hypothesis strictly by its pre-registered criteria. One decision per hypothesis per week — no daily peeking for promotion decisions (daily ops monitoring is separate).
4. **New hypotheses:** pre-register BEFORE computing outcomes; must state the mechanism (who is on the other side, why does the effect persist). Cap: 10 active.
5. **Ship:** approved live changes deploy Monday. Update the decision log.

Active register (max 10)

Hypothesis	Gate	Next checkpoint	Kill criteria
Asia HMM-{H2,H5}-gate	KILLED Aug 1	—	OOS avg -1.92 on $n=15$ (extended); see decision log
4/6/8 ATR stop variants (Asia + full-session NY)	G0 (first read done; not G1)	Aug 9 NY day re-count; Aug 30 G2	no variant beats 10 ATR with positive OOS by Aug 30; NY still <5 live-like days $\rightarrow$ extend
Bull Asia short	provisional live at 4%; formal G1 $\rightarrow$ G2	current bull/neutral window review (Asia live-like Jul20-30 still red)	negative in next / current multi-day bull window
NY quality floor (follow_3h_all n session=ny vs ny_flush_buy_4h)	G0	after ≥ 15 post-fix accepted OOS; Aug review	$< +0.5$ ATR improvement vs paired baseline, or candidate $< +1.0$, or day-concentrated
London follow_3h_london + Late fade_6h_late expansion	G0; quality floor active	Aug 9 earliest if ≥ 20 fresh OOS	concentration fails; or no fills for 2 weeks $\rightarrow$ park
1h time-exit variants (ny_flush_buy_1h, asia_pump_short_1h, t12)	G0	after ≥ 15 post-fix accepted OOS each	$<$ baseline 4h paired Δ avg, or day-concentrated
Regime age gate (max_regime_ageBars; neutral 24-48h NY toxic)	G0 (plumbing live, gate inert)	Aug 23 read; enable only after OOS confirm	toxic cell not reproduced OOS, or gate cuts valid winners (paired $\Delta \leq 0$)
NY scale-in $+0.5$ ATR @ 1h	G0	Aug 16-23 (needs $n \geq 30$)	scale-in net $\leq$ plain single-entry, or day-concentrated
Asia/NY weekend tradability	measurement $\rightarrow$ G0	Aug 16 read; Sep 6 verdict	weekend paired Δ avg ≤ 0 vs weekday, or concentration fails
Funding-rate / OI-delta conditioning	G0	Aug 16 (≥ 3 wk enriched fields)	conditioning adds $< +0.5$ ATR vs baseline, or unstable sign across weeks
Cluster breadth / market-wide liq flow filter	G0	Sep 6 checkpoint (after candidate cap-3 trusted)	no incremental edge vs single-symbol cap-3, or concentration fails

Open register slot: 0 free (10 active). Candidates only via pre-registration before outcome compute. Do not auto-fill — a new candidate requires killing or promoting one.

Measurement / backlog (not active promotion)

Item	Status	Notes
Age throttle (4% if neutral $> 64h$)	live	weekly; remove if $>64h$ cells not worse 3 weeks
Regime-detector audit	measurement	14k+ snapshots in v6_telemetry.db; report lag/flips/transition-zone PnL; do not retune on same window
Asia partial 50% @ $+2$ ATR at 1h	measurement	week of Jul 27 OOS check
Asia limit-entry (-1.5 ATR)	already shadow-logging	research-only until live limit path exists
Symbol allowlist / fee micro / MAE-peak exits	ignore for now	noise / non-executable as tested
TSL / 24h hold enablement	killed / ignore	plain time exits win on current books
Per-regime strategy maps (engine)	engineering backlog	build only after ≥ 2 session $\times$ regime cells pass G2 and require different strategies/stops

Weekend data status

Resolved Jul 22: live Asia and NY both exclude weekends. NY also excludes Mondays. Shadow continues for measurement. Historical Late weekend +3.37 (n=37) is pre-promotion shadow evidence only; quality-floor Late has had no fills since ~Jul 16 — OOS clock has not started.

Backlog (unscheduled — pre-register before touching)

- Symbol heterogeneity / whitelists (needs per-symbol $n \geq 30$).
- Proper MDP sizing with portfolio state — only after ≥ 2 months of live PnL.
- HMM state/confidence as live-logged columns — only if HMM gate reaches G3.
- Per-regime live engine maps — gated on dual G2 cells requiring different rules.
- London/Late half-risk live pilot — only after G2 on fresh quality-floor OOS.

Decision log (do not relitigate without new data)

- **Jul 15 — TSL variants: killed.** Plain time exits beat TSL on both live books.
- **Jul 15 — NY Mondays: blackout.** -48 ATR on Monday samples.
- **Jul 18 — Stops stay 10 ATR live.** 2-5 ATR stops kill valid Asia winners (winner MAE p95 = 5.2 ATR); revisit only via shadow stop ladder + Aug checkpoints.
- **Jul 18 — No scale-ins live.** Asia scale-in negative; NY promising but thin.
- **Jul 18 — Session carryover: dead.** No significant Asia→London→NY PnL link.
- **Jul 18 — MDP on (regime, age) alone: parked.** Degenerate without portfolio state.
- **Jul 19 — Weekends: unproven.** Asia weekend negative vs weekday; prompted Jul 22 exclusion.
- **Jul 22 — Asia session/regime:** keep `asia_pump_short_4h` at 48 bars; allow neutral+bull, drop thin bear, exclude Sat/Sun. Reject blind 36-bar cut.
- **Jul 22 — NY session/regime:** full-session `ny_flush_buy_4h`; neutral+bear only; exclude Mon/Sat/Sun. NY bull remains out.
- **Jul 22 — Late/London:** not live. Matrices are hypothesis generators until fresh quality-floor OOS exists.
- **Jul 23 — Fill accounting:** resolve fills via `userTrades`; refuse unresolved entry/exit prices; net PnL/R includes commissions. Weekly cooldown removed.
- **Jul 23 — Cap-3 integrity:** `would_live_accept` is per-strategy from this date. Pre-Jul-23 shared-column labels are invalid for promotion.
- **Jul 23 — Ignore for now:** symbol allowlists, fee micro-tuning, MAE-peak exits, enabling TSL/24h. Next dollar of research effort: NY quality floor OOS + stop ladder day-count; growth bets (London/Late) wait for fills; per-regime engine waits for dual G2 cells.
- **Aug 1 — Asia HMM {H2,H5} gate: KILLED.** Frozen train $\leq$ Jun30. Primary Jul18-24 EXTEND (in_cap3=11<15, avg=-2.76). Extended Jul18-31 KILL (in_cap3=15, avg=-1.92, blocked_sum=-3.22). Jul21 single-day -30.6 ATR drives H5 OOS. Selection Jul1-17 avg=+1.81 did not transfer. No G3 pilot. No Aug1 gate retrain. NY H4+H5 still useless (n=3). Artifacts under `research/output/reports/hmm_oos_*_2026-08-01.*`.
- **Aug 1 — Stop ladder first read: no live change.** Live-like books all negative in window; relative winner s4 (Asia paired Δ avg +0.38, NY +0.21) via more stops, not a green edge. s6/s8 no clear win. NY only 4 live-like days (need ≥ 5). Keep 10 ATR live through Aug 30 G2 gate. CSVs: `stop_variant_*_2026-08-01.csv`.
- **Aug 16 — Asia regime filter: neutral-only.** Drop bull from live `allowed_btc_regimes` (`live_burst_ny_asia.yaml`). Bull-Asia insufficient evidence (n=142 blocked live, regime frequency low); neutral carries the edge. Revisit on next multi-day bull window.
- **Aug 16 — NY hours pilot: [16,17].** Shift live NY window from [14,15] to [16,17] (`live_burst_ny_asia.yaml`). Variance scan shows h16-17 edge; full-session reverted. Revisit after OOS accumulation.
- **Aug 16 — Regime age gate: plumbing added (inert).** Added `max_regime_age_bars` to `SessionBurstRule` + `LiqBurstFollowConfig`, engine age gate, and `compute_regime_age_bars` wiring in `main.py`. Default None = no-op, so live behaviour unchanged until explicitly enabled. Motivation: variance scan toxic cell neutral 24-48h NY -0.75 R (n=623). Enable only behind a pre-registered threshold
- OOS confirmation.

- **Aug 16 — 1h time-exit variants registered (G0).** Added `ny_flush_buy_1h` and `asia_pump_short_1h` (`timeBars=12`) to shadow. Motivation: ~70% of winners peak by bar 6 (1.5h); 48-bar (4h) exit holds too long. Shadow-only until G0 read.